Math Modelling Homework 1

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1 The SI model as a special case

Prompt: Set $\gamma = 0$ (no recovery) in the SIR system from Day 1. Reduce the system to the two-compartment SI model and show that I(t)→N0 as t→∞ whenever $I0 > 0$. Interpret this biologically.

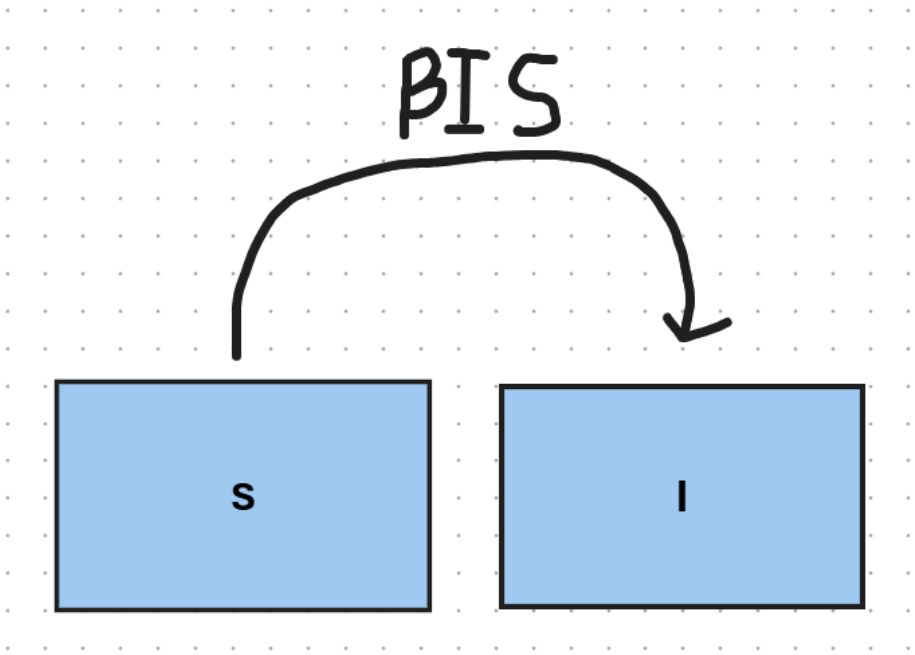


Figure 1: SI Model diagram

$$\frac{dS}{dt} = -\beta * I * S \tag{1}$$

$$\frac{dI}{dt} = \beta * I * S \tag{2}$$

Because there is no removal from the infected category, and there is no way for any subjects to return to the susceptible category, if there is even a single infected subject, eventually all subjects will be infected.

2 Threshold in a concrete case

Prompt: Suppose $\beta = 0.0005\ person^{-1}\ day^{-1}$, $\gamma = 0.2\ day^{-1}$ and $N_0 = 1000$ with $S_0 = 990$. Compute R_0. Does an epidemic occur? At what value of S_0 (holding β, γ fixed) does the population sit exactly at threshold?

$$R_0 = \frac{\beta S_0}{\gamma} = \frac{0.0005 * 990}{0.2} = 2.475 \tag{3}$$

An epidemic occurs, because $R_0 > 1$. The population sits exactly at the threshold at:

$$S_0 = \frac{\gamma}{\beta} = \frac{0.2}{0.0005} = 400 \tag{4}$$

3 Modeling Cycle Short-Answer

Prompt: Pick one of the seven Kermack–McKendrick assumptions from Day 1 and describe a real disease for which it would be a poor approximation, and what modification (structurally, not necessarily solved) you would make to the model.

Assumption 4, that anyone who is infected is immediately infectious, is violated by any disease with an incubation period, such as chickenpox, which has an incubation period of fourteen to sixteen days. In order to account for this, I would add a section between Susceptible and Infected that once a Susceptible person would be infected, they are moved to the Developing box, where they stay for the incubation period and leave after they become infectious. This slows the rate that models grow, because the newly infected won't become infectious for the duration of the incubation time.

4 Reading Check

Prompt: In 3–4 sentences, explain the difference between incidence and prevalence (see Day 1, "The compartmental approach"), and give an example of a public-health question that requires one but not the other.

Incidence is the rate at which new infections occur. Prevalence is the number of currently infected people. Incidence is important for measuring the way the disease is spreading, while prevalence is more relevant for measuring the burden

on hospitals and schools by measuring how many people are capable of spreading the disease or how many are in need of treatment.